MIL-C-10857D(EA)
12 November 1974
SUPERSEDING
MIL-C-10857C(MU)
20 January 1970

MILITARY SPECIFICATION

CLOTH, ANTIFOGGING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of lens antifogging cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-504 - Plastic Sheet and Film, Cellulose Acetate.
CCC-C-458 - Cloth, Cotton, Flannel.

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).
FED-STD-595 - Colors.
FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 4240

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DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

A5-16-2	-	Cloth, Antifogging.
DLB136-10-150	-	Test Device, Fogging, Q176, Data List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The antifogging cloth shall consist of a base fabric conforming to type III, class 1, of CCC-C-458 impregnated with any nontoxic surface-active agent that permits the cloth to conform to this specification.

3.2 Dimensions. The dimensions of the antifogging cloth shall be 6.00 inches ± 0.25 inch by 10.00 inches ± 0.25 inch (see Drawing A5-16-2).

3.3 Edging. All edging of the cloth shall be pinked or free from raveled edges.

3.4 Odor. The antifogging cloth shall have no objectionable odor.

3.5 Color. The color of the antifogging cloth shall approximate color gray No. 36307 of FED-STD-595.

3.6 Breaking strength. When tested as specified in 4.4.4.1, the antifogging cloth shall have a breaking strength, before aging, of no less than 90 percent of that required for the base fabric. After aging as specified in 4.4.4.5, the cloth shall have a breaking strength of no less than 67.5 percent of that required of the base fabric.

3.7 Toxicity. The antifogging cloth shall be nontoxic. The supplier shall submit objective evidence in accordance with 4.1.2.2.

3.8 Mildewproofness. The antifogging cloth shall show no growth of mildew when tested as specified in 4.4.4.2.

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3.9 Antifogging properties. The antifogging properties of the cloth shall be such as to permit the cloth to pass the test specified in 4.4.4.3 both before and after aging as specified in 4.4.4.5.

3.10 Abrasion. The antifogging cloth shall not scratch or abrade cellulose acetate when tested as specified in 4.4.4.4.

3.11 Preproduction. Prior to the start of regular production, a preproduction sample of antifogging cloths shall be produced in accordance with this specification for examination and tests (see 4.3).

3.12 Workmanship. The antifogging cloth shall be free from loose strips, ravelings, damage, and contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence.

4.1.2.1 Materials (3.1) and preparation for delivery (5). The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.1.2.2 Toxicity. The supplier shall submit to the contracting officer within 30 days after date of award of a contract, at no additional cost to the Government, a complete list of the ingredients to be used in formulating the antifogging compound, the percentage of each, and the tolerance that will be permitted. The supplier shall also submit with the above list complete evidence substantiating his claim, to the satisfaction of the Government, that the listed ingredients will be nonirritating or nonsensitizing to the user. This evidence may be in the form of a complete description of the programs that the supplier has carried out, such as patch testing or use testing. He may submit

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evidence that the proposed formulation is essentially the same as that used in his commercial products, and include information relative to the extent and length of time of his commercial experience with the formulation. Other evidence may include reference to technical or other scientific publications, laboratory reports, opinions, or statements of recognized authorities or agencies in the field of dermatology. The Surgeon General will act as advisor in evaluating this evidence for determining compliance.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 100 antifogging cloths shall be produced using the same materials, equipment, processes, and procedures as will be used for regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The 100 antifogging cloths shall be examined in accordance with the classification of defects (4.4.3.3).

4.3.2.2 For tests. Cloths shall be selected at random from the preproduction sample and tested in accordance with table I.

Table I. Preproduction testing.

Requirement	Number of cloths		Test method
	Before aging	After aging	
Breaking strength	5	5	4.4.4.1
Mildewproofness	5	-	4.4.4.2
Antifogging properties	5	5	4.4.4.3
Abrasion	10	-	4.4.4.4

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4.3.3 Acceptance/rejection criteria. If any cloth in the preproduction sample fails to meet all the requirements of this specification then the preproduction sample shall be rejected. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4. Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the antifogging cloths impregnated from one batch of surface-active agent (see 6.3).

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Thirty antifogging cloths shall be selected at random from each lot.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample antifogging cloths shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For test. The sample cloths shall be divided into groups and tested in accordance with table II. The acceptance number is zero.

Table II. Regular production testing.

Requirement	Number of cloths		Test method
	Before aging	After aging	
Breaking strength	5	5	4.4.4.1
Antifogging properties	5	5	4.4.4.3
Abrasion	10		4.4.4.4

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4.4.3.3 Classification of defects.

(a) Antifogging cloth (Dwg. A5-16-2).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Color incorrect
102	Odor objectionable
103	Edges raveled
104	Workmanship (3.12)
Minor:	AQL 6.0 percent defective
201	Dimension incorrect

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Breaking strength. Determine the breaking strength in accordance with method 5100 of FED-STD-191.

4.4.4.2 Mildewproofness. Determine the mildewproofness in accordance with method 5750 of FED-STD-191.

4.4.4.3 Antifogging properties.

4.4.4.3.1 Test device. The fogging test device shall conform to Drawing D136-10-150. The water pan in the warm chamber shall be filled with sufficient water to contain the heating element. The heating element shall be of sufficient size to maintain the temperature of the warm chamber at 95° ± 5° F and the relative humidity at 80 percent ± 5 percent. The cold chamber shall be maintained at a temperature of 30° to 45° F by the use of dry ice placed in the dry ice pan.

4.4.4.3.2 Procedure. The surfaces of 100 sheets of window glass, four inches by four inches by approximately 1/8 inch thick, shall be wiped with the piece of antifogging cloth to be tested. Before the sheets of glass are wiped, they shall be wet uniformly over the surface, as follows: Place two drops of water on one surface of each sheet of glass and spread evenly with the fingers over the whole surface. Each sheet of glass shall be wiped with the same antifogging cloth until the surfaces are dry and clear. After the 100 sheets have been wiped, the surfaces of

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two more sheets of glass similar to the 100, which have been prepared in a similar manner, shall be wiped with the same piece of cloth, until dry and clear, and then placed in the sample slots of the test box with the treated side exposed to the warm side of the test box. The sheets shall be so placed after the air streams are flowing and shall be exposed to the air streams for 15 minutes. During this time, the glass sheets shall be observed for collection of fog or droplets of water by reading through the viewing window the typewritten material fastened on the back wall of the cold chamber of the box. The forming of fog or water droplets to such an extent as to prevent reading of the typed material through more than 25 percent of the area of the glass sheets at any time during the 15 minute test period, shall be considered a failure of the antifogging cloth.

4.4.4.4 Abrasion. The surface of a four-inch by four-inch piece of 1/64-inch thick cellulose acetate sheeting conforming to type 1 of L-P-504 shall be wiped 10 times with each antifogging cloth and visually observed for scratches and abrasion.

4.4.4.5 Properties after accelerated aging. The antifogging cloths to be aged shall be placed in a standard oxygen bomb at 70° ± 2° C and oxygen pressure of 300 ± 5 pounds per square inch (gage) for 72 hours. After aging, the cloth shall be subjected to breaking strength and antifogging tests as specified in 4.4.4.1 and 4.4.4.3, respectively.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking, interplant shipment (see 6.4). The antifogging cloth shall be packaged and packed to provide adequate protection from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with FED-STD-123 and shall include the contractors name, address, and contract number.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the treatment of chemical-biological mask lenses to prevent fogging.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

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(b) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Batch. A batch is defined as that quantity of material which has been subjected to some chemical or physical mixing process intended to make the final product substantially uniform.

6.4 Interplant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor or a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

6.5 International interest. Certain provisions of this specification are the subject of international standardization CSTAG 216 and 234. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization officer, if required.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A635

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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